Unit 2 Poster — Claim, Proof, Repair

Period B · Tue 11/10 + Thu 11/12 build, Fri 11/13 gallery walk · CED topics 2.1 to 2.9 (nothing binomial) · one DOK-3 claim per group

Printables (PDF): group sheets (six cards, two pages each; print one per group) · example board (project this first) · gallery-walk sheet (one per student, Friday) · partner-check sheet (one per group, Thursday) · partner worksheets (each group gets a clean page 1 of its partner's card) · the table and the four claims (for the screen) · teacher key (everything here, with the answers).

The hook for students: "Every claim on these cards is one somebody could really make about data we have already used. Three of them come straight from questions this class asked in the last two weeks. Your group's job is a verdict, the proof, and a repair. Then your poster teaches the room."

The six cards

#	Topics	The claim being judged	The mistake it busts (Period B evidence)	Data
1	2.1, 2.2	"Freeze Time people want to be rich: 26 chose Rich, more than any other power."	"The bigger count wins" **PERSISTENT B**: Period B's Unit 1 error, and 3 students read a relative-frequency bar chart as counts on quiz 2.1 (10/29).	Super Status table, 433 students
2	2.6	"Most students who want to be Famous chose Fly, so most students who chose Fly want to be Famous."	P(A | B) read as P(B | A). Asked in class 10/30: "why do the two conditional probabilities differ?"	Super Status table
3	2.5, 2.7	"Nobody who chose Telepathy wants to be Famous (0 of 68), so Telepathy and Famous are independent."	Mutually exclusive confused with independent. Asked 10/29; you promised the class an example of mutually exclusive events that are not independent. **PERSISTENT B**	Super Status table + a 2x2 table the group designs
4	2.4, 2.7	"P(college grad or internet news) = 693/2500 + 687/2500 = 55.2%, and grads are no more likely than anyone else to get news online: only 9.8% are grads who read news online."	Addition rule without the subtraction; a joint probability used where a conditional belongs. 2.7 is B's least-finished lesson (worksheet mean 46; 8 students not done).	2010 AP exam table, 2,500 adults (2.7 follow-along)
5	2.3, 2.8, 2.9	"A fair game! You win on 2 of every 6 rolls."	Expected value read as "every game"; winnings used instead of net; "fair" without asking for whom (2.8 exit-ticket traps). Also the 10/30 TI-84 promise.	Carnival die game (2.8 follow-along exit ticket)
6 spare	2.3	"6 heads in a row in 20 flips is impossible for a fair coin. It's rigged."	A simulation estimates; it never proves one outcome impossible (10/26: "was the simulated coin fair?").	randInt(0,1,20) on the TI-84

Coverage: 2.1 and 2.2 on card 1; 2.3 on cards 5 and 6; 2.4 everywhere (probabilities as relative frequencies, the complement region on card 4); 2.5 on card 3; 2.6 on cards 1, 2 and 4; 2.7 on cards 3 and 4; 2.8 and 2.9 on card 5. The walk sheet makes every student find one number at each poster except their own. **Period B ran cards 1, 2, 3 and 6.** Unions (the double count) appear only in Thursday's Do Now and expected value not at all: cards 4 and 5 hold both, for Friday's early finishers or for review.

Unit 2 Poster — groups, roles, days, materials

Groups

Class size	Groups	Who has what
13 (Period B, as it ran 11/10)	three trios and one group of four	Cards 1, 2, 3 and 6; the group of four has card 3. Nobody took cards 4 and 5: they are Friday's early-finisher task.

Partner groups for Thursday: poster 2 with poster 3, poster 1 with poster 6.

Roles inside a trio

The whole group judges the claim together **first**, on scratch paper and whiteboards, before anyone touches the poster. Then:

- **Table Keeper** — every fraction and percent on the poster, with the arithmetic shown and the denominator named.
- **Visual** — the diagram the card asks for (segmented bar chart, shaded tables, Venn, 2x2 table, dotplot): title, labels, a key or scale.
- **Writer** — the verdict sentence, the repaired claim and the explanation sentences, in context, using the words from the language objective; checks the must-have list before the walk.

In the group of four there are two Visuals: one draws the shaded table, one the 2x2 table.

The three days (details on the run-of-show pages)

Tue 11/10 · 65 min · solve	Thu 11/12 · 65 min · observed: sketch, check, build	Fri 11/13 · 55 min · finish, gallery walk
Do Now on boards. Objectives + EQ posted. Three Reads on the shared table. Example board. Cards out. Solve the card on scratch paper and whiteboards; write a one-sentence verdict. Exit ticket 1.	Do Now on boards. Sketch on page 2: the mockup. Partner check (poster 2 with 3, poster 1 with 6): work their page 1, look at their mockup, fix your own. Five minutes on the 10/29 promise. Only checked work goes on the big paper. Exit ticket 2, alone, on paper.	Finish against the checklist; partner groups confirm the change. Posters up. Walk sheet at the three other posters. Debrief. Score with the grid.

Materials and grading

Four poster boards and a spare, markers (four colors per group), rulers, tape, scratch paper, a whiteboard + marker + eraser for every student, a TI-84 at every table and three at the poster 1 table (it runs poster 6's simulation), the printed cards (two pages, one per group), a clean page 1 of its partner's card for each group, one partner-check sheet per group (four), 13 walk sheets, 13 of each exit ticket, the hint cards, the objectives on poster paper, the Super Status table on the screen. Each card's five must-haves are the rubric: all five = E, three or four = P, fewer = I. Scores E 115 / P 100 / I 90, any integer 90 to 115 per sheet; enter them with node `scripts/enter-poster.mjs` (format on the rubric page). Posters are 30% of Q2 Work for Period B.

AP Statistics · Period B · Unit 2 Poster

Tue 11/10 · Thu 11/12 · Fri 11/13 **DOK 3** judge it · prove it · repair it

Content objective

I can judge a claim about chance: find the number it really depends on, prove my verdict with it, and repair the claim so it is true.

Language objective

I can defend my verdict out loud and in writing, in complete sentences, using the word my claim needs: **given, union, independent, mutually exclusive, expected value** or **simulation**.

Essential question

What makes a claim about chance worth believing?

Frames

Verdict: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."
Partner check. Checkers write: "We got _____ out of _____. You got _____ out of _____." "Your checklist still needs _____." "A reader on Friday would ask: _____?"
Authors answer: "We will change / keep _____ because _____."

Key words: given · union · independent · mutually exclusive · expected value · simulation **Helpers:** denominator · estimate · long run

This week

- **Tuesday:** verdict and proof on scratch paper and whiteboards.
- **Thursday:** sketch it. Check it with your partner group. Build it.
- **Friday:** finish. Gallery walk: one number at each poster.

Poster 1 — Who wants to be rich?

DOK 3

Data

Super Status! 433 high school students answered two questions: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

The student newsletter wrote: "Freeze Time people are the ones who want to be rich. 26 of them chose Rich, more than any other superpower."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. For each superpower, what percent of its fans chose Rich? Write a fraction and a percent. Which total is the denominator, and why?

> **Answer.** Fly 22/89 = 24.7% · Freeze Time 26/101 = 25.7% · Invisibility 22/98 = 22.4% · Super Strength 13/77 = 16.9% · Telepathy 19/68 = 27.9%. The denominator is the row total: "given Fly" means only the 89 Fly students count. Overall, P(Rich) = 102/433 = 23.6%.

b. Draw a segmented bar chart: one 100% bar per superpower, split into Famous / Happy / Healthy / Rich. Whole percents are fine. Add a key and a percent axis.

> **Answer** (percents within each superpower: Famous / Happy / Healthy / Rich). Fly 5.6 / 53.9 / 15.7 / 24.7 · Freeze Time 3.0 / 62.4 / 8.9 / 25.7 · Invisibility 3.1 / 63.3 / 11.2 / 22.4 · Super Strength 5.2 / 55.8 / 22.1 / 16.9 · Telepathy 0.0 / 72.1 / 0.0 / 27.9. Overall 3.5 / 61.2 / 11.8 / 23.6.

c. Verdict. Do Freeze Time fans stand out for wanting to be rich? Compare their percent with the other four groups and with all 433 students.

> **Answer.** No. The count is right (26 is the largest Rich count), but Freeze Time is also the largest group (101 fans), so a large count is expected. By percent Freeze Time is 25.7%, close to the 23.6% of all students. Telepathy is highest at 27.9% (Freeze Time is second, 2.2 points behind), and every group sits between 17% and 28%. No group stands out: not supported. Also accept the true reading of the count: of the 102 students who want Rich, 26 (25.5%) chose Freeze Time, more than any other power. That is P(Freeze Time | Rich), not the claim's P(Rich | Freeze Time).

d. Repair. Rewrite the claim so it is true. Use a percent, not a count.

> **Answer** (any true version). "About 1 in 4 students in every superpower group wants to be rich (17% to 28%); Freeze Time fans, at 25.7%, are close to the 23.6% of all students." Also true: "Among students who chose Telepathy, 27.9% want to be rich, the highest share; Freeze Time is second at 25.7%."

e. Is there an association between superpower and status? Compare two bars that look different (try Telepathy and Fly) and explain in one sentence.

> **Answer.** Yes: the conditional distributions differ. Telepathy fans: 0% Famous, 0% Healthy, 72% Happy. Fly fans: 5.6% Famous, 15.7% Healthy, 54% Happy. With no association every bar would be split the same way.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 1 — checklist, roles, sketch

Poster checklist (this is the rubric)

- ☐ Five "Rich" percents, each with its fraction
- ☐ Segmented bar chart: five bars, each 100%, four segments, a key, a percent axis
- ☐ A verdict that compares Freeze Time's percent with the other groups and with all 433 students
- ☐ The repaired claim, true, with a percent
- ☐ One association sentence comparing two superpowers' bars

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The five Rich fractions and percents, then the percents for the bars.	The segmented bar chart: five 100% bars, four segments, key, percent axis, title.	The verdict, the repaired claim, the association sentence.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: given · denominator · association

Draft the poster sentences here. Every sentence names the context and uses a number from the table.

Poster 2 — The flipped headline

DOK 3

Data

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

A headline in the school paper: "Most students who want to be Famous chose Fly. So most students who chose Fly want to be Famous."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Find P(Fly | Famous) and P(Famous | Fly) as fractions and percents. Which total is the denominator each time, and why?

> **Answer.** P(Fly | Famous) = 5/15 = 33.3%: the 15 students who want to be Famous (a column total). P(Famous | Fly) = 5/89 = 5.6%: the 89 students who chose Fly (a row total). Same numerator, different "given" group.

b. Shade the "given" group on two copies of the table, one for each probability. Circle the cell the two share.

> **Answer.** Copy 1: the Famous column shaded (total 15) for P(Fly | Famous). Copy 2: the Fly row shaded (total 89) for P(Famous | Fly). The shared cell is 5: same numerator, different given group.

c. Verdict on each half. Is "most" right in the first sentence? Does the second sentence follow from the first?

> **Answer.** First half: Fly is the top pick among the 15 Famous students (5; next is Super Strength with 4), but 33% is not "most": a plurality, not a majority. Second half: it does not follow; the condition is flipped. Only 5.6% of Fly-choosers want to be Famous.

d. Repair. Rewrite the headline as two true sentences, each with its probability.

> **Answer.** "Among the 15 students who want to be Famous, Fly was the most popular power (5, or 33%). Among the 89 who chose Fly, only 5.6% want to be Famous."

e. Build. Find another pair in the table where flipping the "given" changes the answer a lot. Write both probabilities and explain in one sentence why they differ.

> **Answer** (any correct pair). P(Happy | Super Strength) = 43/77 = 55.8% vs P(Super Strength | Happy) = 43/265 = 16.2%. P(Happy | Freeze Time) = 63/101 = 62.4% vs P(Freeze Time | Happy) = 63/265 = 23.8%. P(Rich | Telepathy) = 19/68 = 27.9% vs P(Telepathy | Rich) = 19/102 = 18.6%. Why: the numerator is the same cell; the denominator is the size of the given group.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 2 — checklist, roles, sketch

Poster checklist (this is the rubric)

- ☐ Both conditional probabilities as fractions and percents
- ☐ Two copies of the table, each with its "given" group shaded
- ☐ A verdict on each half of the headline, each with its number
- ☐ The repaired headline: two true sentences with their numbers
- ☐ A second flipped pair from the table with both probabilities and a one-sentence why

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The two fractions and percents, and the second flipped pair. Name each denominator.	The two shaded tables, with the shared cell circled.	The verdict on each half, the repaired headline, the why sentence.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: given denominator

Draft the poster sentences here.

Poster 3 — Can't happen together, so unrelated?

DOK 3

Data

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

"*Nobody who chose Telepathy wants to be Famous: 0 of 68. Since the two never happen together, Telepathy and Famous are independent. One has nothing to do with the other.*"

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Are "chose Telepathy" and "wants to be Famous" mutually exclusive? Show the joint probability P(Telepathy and Famous).

> **Answer.** P(Telepathy and Famous) = 0/433 = 0. No student in this table did both, so for a student picked from this table the two events are mutually exclusive. (The 0 shows it did not happen among these 433, not that it never could.)

b. Are they independent? Compare P(Famous) with P(Famous | Telepathy).

> **Answer.** P(Famous) = 15/433 = 3.5% but P(Famous | Telepathy) = 0/68 = 0%: not equal, so not independent. (The product check agrees: P(Famous) × P(Telepathy) = (15/433)(68/433) = 0.0054, but P(Famous and Telepathy) = 0.)

c. Verdict, in one sentence that uses both terms correctly. What does knowing "Telepathy" do to the chance of "Famous"?

> **Answer.** "Telepathy and Famous are mutually exclusive but not independent: knowing a student chose Telepathy drops the chance they want to be Famous from 3.5% to 0." Seeing one rules the other out, which changes its probability: that is dependence, not independence.

d. Prove it in general. Two events can each happen, and they are mutually exclusive. Explain why they can never be independent. (What does independence say P(A and B) must equal?)

> **Answer.** Independent means P(A and B) = P(A) × P(B). If both events are possible, that product is above 0. Mutually exclusive means P(A and B) = 0. Both cannot be true at once, so mutually exclusive events with positive probabilities are never independent.

e. Build. Make a 2x2 table of 100 students for "plays a sport" and "plays an instrument" where the two events are independent but NOT mutually exclusive. Write the check under it.

> **Answer** (one of many). Independent: sport 60, instrument 50, both 30 → sport only 30, instrument only 20, neither 20. Check: P(instrument | sport) = 30/60 = 50% = P(instrument) = 50/100. Any table with both = P(S) × P(I) × 100 works. Extension for a fast group: a second table where the events ARE mutually exclusive (sport 50, instrument 50, both 0, neither 0); the same check fails, 0/50 = 0% against P(instrument) = 50%.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 3 — checklist, roles, sketch

Can't happen together, so unrelated?

Poster checklist (this is the rubric)

- ☐ The joint probability P(Telepathy and Famous) from the table, and what it tells you
- ☐ The independence test, with its two numbers
- ☐ One verdict sentence that uses *mutually exclusive* AND *independent* correctly
- ☐ The proof from part (d), in two or three sentences
- ☐ Your 2x2 table (100 students) with the independence check written under it

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The probabilities, and the numbers in your 2x2 table.	The big table with the Telepathy row and the Famous column shaded, and your 2x2 table with its check underneath.	The verdict sentence and the proof.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: mutually exclusive | independent | given

Draft the poster sentences here.

Poster 4 — Double counted

DOK 3

Data

A survey of 2,500 adults asked about education level and where they get their news. One of these 2,500 adults is picked at random.

Group	Count
College graduates	693
Get news mainly from the internet	687
College graduates who get news from the internet	245
All adults surveyed	**2,500**

The claim you are judging

A blog wrote: "P(college grad or internet news) = 693/2500 + 687/2500 = 55.2%. And college grads are no more likely than anyone else to get their news online: only 245/2500 = 9.8% of adults are grads who read news online."

Two mistakes are hiding in this claim. Find both.

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Draw a Venn diagram (or a two-way table) and fill in all four regions: both, grad only, internet only, neither. Check that they add to 2,500.

> **Answer.** Both 245 · grad only 693 − 245 = 448 · internet only 687 − 245 = 442 · neither 2,500 − 1,135 = 1,365. Sum 2,500.

b. Find the mistake in the 55.2%. Compute the correct P(grad or internet). Which people were counted twice?

> **Answer.** The 245 adults who are both were counted in 693 and again in 687. P(grad or internet) = (693 + 687 − 245)/2500 = 1135/2500 = 45.4%. (The blog's 1380/2500 = 55.2% is a union with double counting.)

c. The blog backs its second sentence with 9.8%. Is 9.8% a joint or a conditional probability? Which probability does "no more likely than anyone else" really depend on?

> **Answer.** 9.8% = 245/2500 = P(grad and internet), a joint probability out of all 2,500 adults. "No more likely than anyone else" compares grads with everyone, so it depends on the conditional P(internet | grad), set beside P(internet).

d. Test the second sentence with the right probability. Are the two events independent? Who is more likely to read news online?

> **Answer.** P(internet | grad) = 245/693 = 35.4% vs P(internet) = 687/2500 = 27.5%: not equal, not independent. Grads are MORE likely to read news online (35.4% vs 27.5% overall, and vs 442/1807 = 24.5% of non-grads). The blog's second sentence is false. (The product check agrees: P(grad) × P(internet) = 0.2772 × 0.2748 = 0.0762 vs P(both) = 245/2500 = 0.098.)

e. Build. Keep 693 college grads and 687 internet readers. How many people in "both" would independence predict? Compare that with 245. What does the gap tell you?

> **Answer.** both = 693 × 687 / 2500 = 190.44 people, about 190. The survey found 245, about 55 more than independence predicts, so the dependence is large, not rounding. (190.44 is not a whole number, so tables of these sizes can never be exactly independent.)

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 4 — checklist, roles, sketch

Double counted

Poster checklist (this is the rubric)

- ☐ Four regions filled and summing to 2,500 (both / grad only / internet only / neither)
- ☐ The people counted twice, named, and the corrected union as a fraction and a percent
- ☐ The blog's 9.8% named (joint or conditional?) and the probability it should have used
- ☐ The independence test with its two numbers, and a verdict naming who is more likely
- ☐ The "both" count independence would predict, compared with 245

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The four regions, the union, the test, the predicted "both" count. Show every subtraction.	The Venn diagram with the overlap shaded (or the 2x2 table), every region labeled with its count.	The double-count sentence, the sentence about the 9.8%, the verdict.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: union · given · independent

Draft the poster sentences here.

Poster 5 — Is the game fair?

DOK 3

Data

A carnival game costs $2 to play. You roll a fair six-sided die. Roll a 6: you win $10. Roll a 1: you win $3. Otherwise: you win nothing.

The claim you are judging

The sign on the booth: "A FAIR GAME! You win on 2 of every 6 rolls."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Let X = the player's net profit for one game (winnings minus the $2). List every value of X with its probability in a table. Check that the probabilities add to 1.

> **Answer.** X = +8 (roll 6) with 1/6 · +1 (roll 1) with 1/6 · −2 (roll 2, 3, 4, 5) with 4/6. Sum 6/6 = 1. Using $10, $3, $0 instead of net is the first trap.

b. Compute the mean μ_X and the standard deviation σ_X. Show the formula and the substitution; a calculator may do the arithmetic.

> **Answer.** $\mu_X = 8(1/6) + 1(1/6) + (-2)(4/6) = (8 + 1 - 8)/6 = 1/6 \approx \$0.17$. $\sigma_X^2 = (8 - 1/6)^2(1/6) + (1 - 1/6)^2(1/6) + (-2 - 1/6)^2(4/6) = 485/36 \approx 13.47$, so $\sigma_X \approx \$3.67$. Shortcut: $E(X^2) = (64 + 1 + 16)/6 = 13.5$, minus $\mu^2 = 1/36$.

c. Interpret μ_X and σ_X in context. Use "in the long run" and say "per game".

> **Answer.** In the long run, over many games, a player gains about 17 cents per game on average. A single game's net typically differs from that average by about $3.67.

d. Verdict. Is the game fair? Who does it favor? Why is "win on 2 of every 6 rolls" the wrong number for judging fairness?

> **Answer.** A fair game has $\mu_X = 0$, for both sides. Here $\mu_X = +\$0.17$, so the game favors the PLAYER: the booth loses about 17 cents per game in the long run. "2 of 6" is the chance of winning something on one roll (not a promise of two wins in six). It counts how OFTEN you win, not how MUCH: the $10 prize outweighs the four $2 losses.

e. Repair. Change ONE payout so that $\mu_X = 0$ exactly. Show the new table and the new mean.

> **Answer** (either). The 6 pays $9 (net +7): $\mu = (7 + 1 - 8)/6 = 0$. Or the 1 pays $2 (net 0): $\mu = (8 + 0 - 8)/6 = 0$. Raising the price to $13/6 \approx \$2.17$ also balances it (exactly only at $13/6$) but is not a payout change.

f. Simulate 20 games. On the TI-84: randInt(1,6,20) STO→ L1, then read the 20 rolls in STAT → EDIT. Use your table to turn each roll into its net, and find the average. Why does it differ from μ_X? What would 1,000 games do?

> **Answer.** Twenty-game averages vary a lot (their standard deviation is about $3.67/\sqrt{20} \approx \$0.82$), so most groups land somewhere between about −$1.50 and +$1.80. With 1,000 games the average settles near $0.17: the law of large numbers.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 5 — checklist, roles, sketch

Poster checklist (this is the rubric)

- ☐ The distribution table of X = net profit, with probabilities that sum to 1
- ☐ μ_X and σ_X, each with its formula and substitution
- ☐ Both interpretations in context, with "in the long run" and "per game"
- ☐ A verdict naming who the game favors and why "2 of 6" is the wrong number
- ☐ The one-payout fix (new mean shown), and a dotplot of your 20 simulated games with their average marked

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The table, μ, σ, the fix. Show the formula before the numbers.	Runs the 20 games on the TI-84. The dotplot of the 20 nets, with their average and μ_X both marked.	The two interpretations and the verdict.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____." **Also:** "In the long run, _____ per _____."

Your words: expected value | long run | net

Draft the poster sentences here.

Poster 6 (spare) — Rigged coin?

DOK 3

Data

You make the data: 20 flips of a fair coin on the TI-84, where 1 = heads. randInt(0,1,20) STO→ L1, then read the flips down L1 in STAT → EDIT. (randInt is under MATH → PROB.)

The claim you are judging

"I flipped a coin 20 times and got 6 heads in a row. That can't happen with a fair coin. It's rigged."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Design. Write the simulation in three sentences: how one flip is modeled, what one trial is, and what you record from each trial (the longest run of heads in the 20 flips).

> **Answer.** One flip: a random digit 0 or 1, 1 = heads, each equally likely. One trial: 20 such digits, randInt(0,1,20). Record: the longest string of consecutive 1s in the trial.

b. Run 25 trials. Record the longest run each time. Draw a dotplot of the 25 values.

> **Answer.** Typical values 3 to 5; a 6 or more appears in roughly 3 of 25 trials.

c. Estimate P(longest run is 6 or more) from your dotplot.

> **Answer.** (number of trials with a run of 6 or more)/25. Expect something near 0.12; anything from 0 to 0.24 is normal variation for 25 trials. If a group gets 0, pool the class's trials before the verdict.

d. Verdict. First decide: how rare would 6 in a row have to be before you believed the coin was rigged? Then compare your estimate with that bar. Is 6 in a row convincing evidence?

> **Answer.** No. A fair coin gives a run of 6 or more heads in about 12% of sets of 20 flips, roughly one set in eight. That is not rare, so it is not convincing evidence of rigging. It does not prove the coin is fair, either.

e. Your estimate is not the exact answer. Why not? What would 1,000 trials do? (Your teacher has the exact value.)

> **Answer.** A simulation gives a relative frequency, which varies from run to run; more trials make it settle closer to the exact value 0.1223 (law of large numbers). For reference: a run of 5 or more, 0.250; 7 or more, 0.058; 8 or more, 0.027.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 6 — checklist, roles, sketch

Poster checklist (this is the rubric)

- ☐ The design in three sentences (flip, trial, what is recorded)
- ☐ 25 trials recorded in a table
- ☐ A dotplot of the 25 longest runs with a scale and a title
- ☐ The estimated probability as a fraction of 25 and a decimal
- ☐ The verdict, plus one sentence on what more trials would do

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
Runs the calculator, records the 25 longest runs, computes the fraction.	The dotplot, with a scale that covers every value you got and the "6 or more" region shaded.	The design sentences and the verdict.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: simulation · trial · estimate

Draft the poster sentences here.

Exit ticket 1 — Tuesday (two per page, cut)

Exit ticket 1 · Tue 11/10 Name _____ Poster #____

In your own words, no copying from the poster:
Our claim says _____.

_____

The number it really depends on is _____ = _____ (fraction, percent, or dollars).
So the claim is (supported / not supported / partly supported) because

_____

_____

_____

Use at least one of: given · union · independent · mutually exclusive · expected value.

- cut here -

Exit ticket 1 · Tue 11/10 Name _____ Poster #____

In your own words, no copying from the poster:
Our claim says _____.

_____

The number it really depends on is _____ = _____ (fraction, percent, or dollars).
So the claim is (supported / not supported / partly supported) because

_____

_____

_____

Use at least one of: given · union · independent · mutually exclusive · expected value.

Exit ticket 2 — Thursday (two per page, cut)

Exit ticket 2 · Thu 11/12 Name _____ Alone, on paper.

1. In the Super Status table, nobody who chose Telepathy wants to be Healthy (0 of 68). A classmate says: "So Healthy and Telepathy are independent." True or false? One sentence why, with a number. (P(Healthy) = 51/433.)

2. A classmate says: "P(Happy | Super Strength) and P(Super Strength | Happy) are equal, because both use the same 43 students." Is the classmate right? Show both as fractions and percents, and say why. (Super Strength total: 77. Happy total: 265.)

- cut here -

Exit ticket 2 · Thu 11/12 Name _____ Alone, on paper.

1. In the Super Status table, nobody who chose Telepathy wants to be Healthy (0 of 68). A classmate says: "So Healthy and Telepathy are independent." True or false? One sentence why, with a number. (P(Healthy) = 51/433.)

2. A classmate says: "P(Happy | Super Strength) and P(Super Strength | Happy) are equal, because both use the same 43 students." Is the classmate right? Show both as fractions and percents, and say why. (Super Strength total: 77. Happy total: 265.)

Run of show, Tuesday

Tue 11/10 · 09:55-11:00 · solve (65 min)

As it ran: 13 students in four groups (posters 1, 2, 3 and 6) solved page 1 of their cards. The sketch and the big paper moved to Thursday.

Min	Students	Teacher
0-5	Do Now on whiteboards: "From the table, P(Rich | Fly)?" Boards up on the cue.	Scan. Ask one board: "Why 89 and not 433?" (22/89 = 24.7%.)
5-13	Three Reads on the projected table. Read 1: what is the story (no numbers). Read 2: list the quantities on your board. Read 3: what claims could someone make from this table?	The objectives, the EQ and the frame are already up on poster paper: point to them before Read 1. Collect three claims from boards; do not judge them yet.
13-17	Watch the example board.	Talk-track (3 min): the word that names the group, the denominator, verdict, repair, seven parts.
17-20	Groups form; cards and materials out; roles chosen and written on the card.	Hand each group its card.
20-38	Solve parts (a)-(c) on scratch paper and whiteboards. No poster yet.	Circulate with the why questions. At minute 28, tier-1 hint card to any blank board; tier-2 only after that.
38-60	Finish (d) and (e) and the one-sentence verdict on page 1. (Planned: start the build. It moved to Thursday.)	Check each group's verdict sentence. Cards stay in the room.
60-65	Exit ticket 1, alone.	Collect. Sort by card for Thursday's partner check.

Run of show, Thursday

Thu 11/12 · 09:55-11:00 · observed: sketch, check, build (65 min)

Partner groups sit at neighboring tables: poster 2 with poster 3, poster 1 with poster 6. Each group gets a clean page 1 of its partner's card: poster 3's page to the poster 2 group, poster 2's to the poster 3 group, poster 6's to the poster 1 group, poster 1's to the poster 6 group. Tier-2 hint cards stay in your hand until the check is over. **If the clock runs over:** exit ticket 2 starts at minute 57 no matter what; the build shrinks first (posters finish Friday). Never cut Work or Look.

Min	Students	Teacher
0-5	Do Now on whiteboards, alone: "True or false: P(Happy or Fly) = 265/433 + 89/433." T or F and one reason. Boards up.	Two why questions, five seconds of wait after each: "Who got counted twice, and how do you know?" "What would make the statement true?" (False: (265 + 89 − 48)/433 = 306/433 = 70.7%.)
5-6	One student reads the language objective aloud.	On the board: Sketch · Check · Build. "On Friday nobody stands beside your poster. Its reader will ask something. Today your partner group is that reader."
6-17	**Sketch.** Page 2, in pencil: the visual in the box, the poster sentences on the lines, the verdict with the frame. This page is your mockup. Poster 6 finishes its trials first and stops at minute 14 wherever it is.	Poster 6 first: a different seed on each calculator (type a number, STO→ rand). One why question at each table, then wait. Minute 14: "Three minutes." Minute 17: "Mockups face down." Then one partner-check sheet and the partner's clean page 1 to each group.
17-37	**Partner check**, the three steps on the sheet. **Work** (9 min): at your own table, solve your partner group's page 1 from the clean copy. **Look** (7 min): swap tables, turn their mockup over, compare it with your work, write on their sheet. **Fix** (4 min): back home; change or keep; the verdict again; tell your partner group your answer.	Call the three steps by the clock. Work first, look second: no mockup is turned over before minute 26. Three TI-84s at the poster 1 table for Work, each seeded first with a different number (type a number, STO→ rand). In Look, stand where posters 2 and 3 are being checked. One why question at a table, then wait; never say a number. A group with no mockup yet: its checkers read the verdict on its page 1.
37-42	Whole class, on whiteboards, alone: "Nobody who chose Telepathy wants to be Famous. Does knowing someone chose Telepathy change the chance they want to be Famous? Yes or no, plus one number." Boards up.	Wait five seconds. One of poster 3's checkers says what they first thought. The poster 3 group gives the numbers: 0/68 against 15/433 = 3.5%. Name the rule (the 10/29 promise): "Can't happen together rules the other one out, which changes its probability: dependence, not independence." No Healthy numbers: that is the exit ticket.
42-56	**Build.** Big paper, by role, from the checked mockup. First pencil a box for each part, so all the roles work at once. Then the second-draft verdict, the numbers with their fractions, the visual, the repair. Title and color last.	Read each second-draft verdict before it is lettered and ask one why question about it, then wait. Say-it check: a student who is not the Writer tells you the verdict with the frame and the key word. Posters are finished Friday.
56-57	Posters and partner-check sheets to the side table; markers capped; desks cleared to a pencil.	Exit ticket 2 goes out face down.
57-65	Exit ticket 2, alone, on paper.	Collect at 11:00: the independent evidence for the day.

Run of show, Friday

Fri 11/13 · 08:55-09:50 · finish, gallery walk (55 min)

Min	Students	Teacher
0-20	Finish the poster against the checklist. At minute 15 partner groups trade places for three minutes: is Thursday's change on the poster? The bottom half of the partner-check sheet is taped to the poster. Posters up around the room.	Hand out walk sheets. A group that finishes early takes card 4 or card 5 and judges the claim on whiteboards: the number and the verdict, no poster.
20-38	Walk: three rotations of 6 min, no docents; the poster must speak for itself (that is what the Writer's sentences are for). Start at the poster to your group's right. At each poster except your own, find the number on the walk sheet. At poster 6, count the dots.	Signal rotations. Listen at posters 2 and 3 (the flipped given; the 10/29 promise).
38-48	Debrief: hands for "which claim seemed true at first?"; two students read their sentence for posters 2 and 3; the poster 6 group says what its checkers' six trials showed.	Name the two rules: the given group is the denominator; mutually exclusive is not independent. Then a third, from poster 6: a small simulation wobbles, and "not rare" is not proof of "fair". Only now give the exact 0.1223.
48-55	The last walk-sheet question, alone: what makes a claim about chance worth believing? Hand in walk sheets.	Score with the grid (or after class); photos of the posters.

Observer card — Thursday 11/12

Lesson	Unit 2 claim posters, day 2 of 3: sketch, check, build (DOK 3)
Plan	**Preparing (6 min).** Do Now on whiteboards, a packet problem: true or false, P(Happy or Fly) = 265/433 + 89/433. Boards up. Two open questions ("Who got counted twice, and how do you know?" "What would make the statement true?"). A student reads the language objective aloud. **Interacting (31 min).** Sketch (11 min): each group drafts its poster and its verdict on the second page of its card. That page is its mockup. Partner check (20 min): poster 2 with poster 3, poster 1 with poster 6. Each group first solves its partner group's claim from a clean copy of that group's card. The groups then swap tables, set the partner's mockup beside their own work, and write what differs, what its checklist still needs and what a reader would ask. Back at their own table the authors decide what to change or keep, write a second draft, and tell their partners why (a one-round form of Stronger and Clearer: first draft, structured feedback, second draft). **Extending (28 min).** Whole class on whiteboards (5 min): does knowing "Telepathy" change the chance of "Famous"? The promised example of events that are mutually exclusive but not independent, with the numbers. Build (14 min): only checked work goes on the big paper, with the second-draft verdict. Clean-up (1 min). Exit ticket 2 alone on paper (8 min): a true-or-false on independence with a number, and a classmate's claim about two conditional probabilities to judge.
Sentence frame	Verdict: "The claim says ___. The number it really depends on is ___ = ___. So the claim is (supported / partly supported / not supported) because ___." Partner check: "We got ___ out of ___. You got ___ out of ___." "Your checklist still needs ___." "A reader on Friday would ask: ___?" "We will change / keep ___ because ___."
Grouping	alone on whiteboards; four groups (three of three, one of four); partner groups for the check; alone for the exit ticket
Minutes	65

What the observer should count: open why questions asked; boards up at each cue; groups whose written feedback names a number or a missing checklist item; verdicts that changed between the first draft and the second; exit tickets collected.

Why questions (open; aim for more than half of all questions)

- Why is the denominator 89 and not 433?
- What would the claim need to say for your number to support it?
- Who got counted twice, and how do you know?
- Why can't these two events be independent if they never happen together?
- How rare is rare enough to call a coin rigged? Who decides?
- Which is the bigger group here, and why does that matter?
- If you only had the percents, could you still tell who is right?
- How would you convince someone who only looks at counts?
- What does the 0 in the table show, and what does it not show?
- Their 6 trials say one thing and your 25 say another. Which do you trust more, and why?
- Can you say that again with the word "given"?
- Which piece of their feedback did you not take, and why?

Where each look-for lives on Thursday

Form item	Where
Objectives posted, content and language	The poster paper, up at 09:55; the language objective read aloud at minute 5.
DOK level	DOK 3 on every card header and on the board: judge a claim, prove it, repair it. In the partner check each group judges a second claim from a clean card, then judges another group's draft against its own work and says what would make it convincing.
Open questions, wait time	The why list; five seconds after each; boards up before anyone answers. In the build, one why question about each second-draft verdict.
Language routine with frames	The partner check: a first draft, one round of structured written feedback from another group, then a second draft (a one-round form of Stronger and Clearer). Frames: the verdict frame, then "We got ___ out of ___. You got ___ out of ___." "Your checklist still needs ___." "A reader on Friday would ask: ___?" "We will change / keep ___ because ___."
Student-to-student discourse	Partner check: nine minutes solving another group's claim together (minutes 17-26), seven minutes judging that group's mockup together and agreeing what to write (26-33), then each group decides what to change or keep and tells its partner group why (33-37).
Whiteboards used	Do Now; the scratch work in the partner check; the Telepathy question.
Visible independent work at the end	Exit ticket 2, alone, on paper, collected.
Scaffolds for multilingual learners	Word bank, sentence frames for the verdict and for feedback, feedback written before it is spoken, tiered hint cards, the data table on every card, roles that give every student a job.
Evidence for feedback and adjustment	The partner-check sheets (the feedback and the second drafts), exit tickets 1 and 2, the walk sheets and the scored posters, sorted by card.

The data and the four claims

For the screen on Thursday: the table for the Do Now and the Telepathy question, and the four claims. Nothing on this page is an answer.

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

Poster 6 makes its own data: 20 flips of a fair coin on the TI-84, where 1 = heads. randInt(0,1,20) STO→ L1, then read the flips down L1 in STAT → EDIT. (randInt is under MATH → PROB.)

Poster 1

The student newsletter wrote: "Freeze Time people are the ones who want to be rich. 26 of them chose Rich, more than any other superpower."

Poster 2

A headline in the school paper: "Most students who want to be Famous chose Fly. So most students who chose Fly want to be Famous."

Poster 3

"Nobody who chose Telepathy wants to be Famous: 0 of 68. Since the two never happen together, Telepathy and Famous are independent. One has nothing to do with the other."

Poster 6

"I flipped a coin 20 times and got 6 heads in a row. That can't happen with a fair coin. It's rigged."

Partner check — before the big paper

Thu 11/12 · one sheet per group

Authors: the group whose claim it is. **Checkers:** the other group. **Checkers ask. Authors decide.** Checkers never write on the authors' card or poster.

WORK (9 min) At your own table. Your mockup stays face down. Solve your partner group's page 1 together, from the clean copy: the number as a fraction, and your verdict. Write the verdict on the clean copy; you do not show it to your teacher.

Checking poster 1 or 2? Do parts (a) and (c). Checking poster 3? Do (a), (b) and (c). Checking poster 6? Do (a), run 2 trials each (6 in all), then (c) from your 6 trials, and (d).

LOOK (7 min) Swap tables. Leave your card and this sheet on your table; bring your work. Turn their mockup over and compare it with your work: the fractions, the verdict, their checklist. One checker writes on the bottom half of the authors' sheet, the one on their table.

FIX (4 min) Go back to your own table. Read what your checkers wrote. Decide: change or keep? Writer: write your verdict again below, stronger and clearer. Then tell your partner group your answer out loud.

Checkers write: "We got _____ out of _____. You got _____ out of _____."
Checkers write: "Your checklist still needs _____." "A reader on Friday would ask: _____?"
Authors answer: "We will change / keep _____ because _____."

- - - - - - - - - - - - cut here: tape the bottom half to your poster - - - - - - - - - - - -

Our poster #____ · checked by the poster #____ group

Checkers write: We got _____ out of _____. You got _____ out of _____.

A second number, if needed: We got _____ out of _____. You got _____ out of _____.

Checkers write: Your checklist still needs

Checkers write: A reader on Friday would ask:

Authors write: We will change / keep (circle one)

because

Our verdict, second draft:

Partner check — what the checkers should find

Why these pairs. Posters 2 and 3 answer the two questions this class asked out loud (10/30 and 10/29). Both turn on the Famous column and the word *given*, so each group already holds the tool the other card needs. Posters 1 and 6 ask the same question, "compared with what?": does 25.7% stand out, is six in a row rare.

Your part. One rule: work first, look second. No group turns a mockup over before it has its own number and verdict. One why question at a table, then wait; never a number. A verdict the checkers dispute changes only after you have heard both groups; otherwise the authors decide in Fix, and you read the second draft in the build.

Poster	Checked by	Their work should show	If it differs from the mockup
1	the poster 6 group: parts (a) and (c)	26/101 = 25.7% of Freeze Time fans chose Rich, against 102/433 = 23.6% of all students; the other rows run from 16.9% to 27.9%.	26/433 = 6.0% and 26/102 = 25.5% are the two traps. 26/102 looks like a match in percents; in fractions it is not. Ask: "Which words in the claim name the group?" A third trap: "25.7% beats 23.6%, so supported." Ask: "Did you check all five rows?" (Telepathy is 19/68 = 27.9%.)
2	the poster 3 group: parts (a) and (c)	First sentence: P(Fly | Famous) = 5/15 = 33.3%. Second sentence: P(Famous | Fly) = 5/89 = 5.6%.	"Fly is the biggest number in the Famous column, so most is right": ask "Most means more than what?" and wait. 5/433 for both sentences: ask "Out of how many, each time?"
3	the poster 2 group: parts (a), (b) and (c)	P(Telepathy and Famous) = 0/433 = 0. P(Famous | Telepathy) = 0/68 = 0 against P(Famous) = 15/433 = 3.5%. The flipped test decides it too: P(Telepathy | Famous) = 0/15 = 0 against P(Telepathy) = 68/433 = 15.7%.	"0 of 68 means no connection, so independent" is the claim itself. Ask: "Does knowing Telepathy change the chance of Famous?" If the two groups still disagree after Fix, that opens the whole-class moment.
6	the poster 1 group: part (a), two trials each, then (c) and (d)	Their own three design sentences (how one flip is modeled, what one trial is, what is recorded) and six trials in all. Six trials show no run of 6 or more about 46% of the time, and exactly one about 38%. In Look they set their design and their six beside the authors' design and 25.	"(1/2)^6 = 1/64, so it is rare": that is six heads on six chosen flips, not a run anywhere in 20. Ask: "Where in 20 flips could a run of 6 start?" If they then say 15 × 1/64 = 23%: the 15 runs overlap, so some sets of flips are counted twice (the Do Now's mistake). Ask: "Can two of those runs happen in the same 20 flips?" "We got 0 of 6, so it can't happen": ask "Your 6 trials or their 25: which do you trust more, and why?"

Calculators. Calculators that were reset together give every student the same flips. Before the first randInt, each student types a different number, then STO→ rand (rand is under MATH → PROB). **Clean copies.** The partner worksheets are page 1 of cards 1, 2, 3 and 6, unmarked: one or two copies for each checking group.

Hint cards — tiered, hand out, never post

Tier 1 goes to a blank board at minute 28 on Tuesday. Tier 2 goes only to a group still stuck after tier 1, and only on the part they are stuck on. Frames come after thinking, not before. On Thursday the tier-2 cards stay in your hand until the partner check is over: they print numbers another group could read.

Tier 1 (any card)

Find the word in the claim that names a group: *given, of, among, who, people*. Circle that group's TOTAL in the table. That total is your denominator. Now find the cell the claim is about and write the fraction.

Tier 2 · Card 1

Freeze Time row: 26 out of 101. Telepathy row: 19 out of 68. Which is bigger as a percent? Do all five rows the same way.

Tier 2 · Card 2

First sentence: you are inside the Famous COLUMN (15 people). Second sentence: inside the Fly ROW (89 people). Same cell, 5, on top of both.

Tier 2 · Card 3

Mutually exclusive asks: can both happen? (Look at the cell.) Independent asks: does knowing one CHANGE the chance of the other? Compare P(Famous) = 15/433 with P(Famous | Telepathy) = 0/68.

Tier 2 · Card 4

Draw two overlapping circles. Put 245 in the overlap FIRST. Grad-only = 693 − 245. Internet-only = 687 − 245. Neither = 2,500 minus everything inside the circles.

Tier 2 · Card 5

Net = winnings − 2. Rolls 2, 3, 4 and 5 all give the same net, so group them: probability 4/6. Mean = each net times its probability, added up.

Tier 2 · Card 6

MATH → PROB → randInt(0,1,20) STO→ L1 puts 20 flips in L1. Read down L1 in STAT → EDIT and count the longest string of 1s. Run it again for the next trial.

Rubric and scoring grid

Each card's five must-haves are the rubric. All five = E = 115. Three or four = P = 100. Fewer = I = 90. Any integer 90 to 115 is allowed per sheet (POSTER_WORK_TRACK_SPEC.md). Posters are 30% of Q2 Work for Period B; a missing poster is absent, never 0.

Card	M1	M2	M3	M4	M5	E / P / I	Score 90-115	Names
1 Who wants to be rich?								
2 The flipped headline								
3 Can't happen together								
4 Double counted								
5 Is the game fair?								
6 Rigged coin?								

Must-haves by card (M1 to M5)

Card	M1	M2	M3	M4	M5
1	Five Rich percents with fractions, row totals as denominators	Segmented bar chart: five 100% bars, four segments, key, percent axis	Verdict compares Freeze Time's percent with the other groups and the overall 23.6%	Repaired claim, true, with a percent	Association sentence comparing two bars
2	Both conditionals, denominators 15 and 89, percents	Two shaded tables showing the given groups	Verdict on each half; "most" vs 50%; the flip named	Repaired headline, two true sentences with numbers	Second flipped pair with both numbers and a why
3	Joint probability 0, "mutually exclusive"	Independence test with numbers (3.5% vs 0)	Verdict using both terms correctly	General argument (ME + possible means not independent)	A 2x2 table, independent but not mutually exclusive, with its check
4	Four regions summing to 2,500	Double count named; corrected union as fraction and percent	The 9.8% named as a joint; the conditional it should have used	Independence test, verdict, who is more likely	Predicted "both" count (about 190) against 245
5	Net distribution table, probabilities sum to 1	μ and σ with formula and substitution	Both interpretations, "long run", "per game"	Verdict: who it favors, why 2 of 6 is the wrong number	One-payout fix with $\mu = 0$, and the 20-game dotplot with its average beside μ
6	Design in three sentences	25 trials recorded (the checkers' six are a separate, smaller run)	Dotplot with scale and title	Estimate as a fraction of the trials run and a decimal	Verdict plus the more-trials sentence

Entering scores and grouping

Entering scores

File outside the repo (student names), then `node scripts/enter-poster.mjs <file>` to preview and `--apply` to write:
`unit=2 quarter=Q2 title=Unit 2 Poster (Claim, Proof, Repair) section=PeriodB`
`Real Name|115` or `Real Name|E` (E 115 / P 100 / I 90), one line per student.

Grouping grid (Period B, as it ran)

Group	Card	Table Keeper	Visual	Writer
A	1			
B	2			
C (four)	3		(two Visuals)	
D	6			

Answer key — Do Nows, exit tickets, the numbers, debrief notes

Do Nows (packet problems, whiteboards)

- **Tue:** P(Rich | Fly) = 22/89 = 24.7%. Denominator 89: inside the Fly row.
- **Thu:** False. P(Happy or Fly) = (265 + 89 − 48)/433 = 306/433 = 70.7%. The 48 students who are Happy AND Fly were counted twice in 354/433 = 81.8%.

Exit ticket 1 (Tue)

Full credit: the claim restated, the right number for that card (a conditional with the right denominator on cards 1-3, the corrected union and the conditional on card 4, the expected net in dollars on card 5, the estimated run probability on card 6), a verdict that matches the card's answer, one word from the bank used correctly. Sort by card; a group whose tickets disagree with each other gets your visit right after poster 6 on Thursday.

Exit ticket 2 (Thu)

1. False. Healthy and Telepathy are mutually exclusive (0 of 68), not independent: P(Healthy) = 51/433 = 11.8% but P(Healthy | Telepathy) = 0/68 = 0%. Knowing Telepathy changes the chance.
2. No. P(Happy | Super Strength) = 43/77 = 55.8% is far bigger than P(Super Strength | Happy) = 43/265 = 16.2%. Same cell (43), different given group (77 vs 265). If the poster 2 group chose this pair as its second example, read their "why" sentence, not their numbers.

The numbers, card by card

Card	Key values
1	P(Rich | power): Fly 22/89 = 24.7%, Freeze Time 26/101 = 25.7%, Invisibility 22/98 = 22.4%, Super Strength 13/77 = 16.9%, Telepathy 19/68 = 27.9%; P(Rich) = 102/433 = 23.6%. All five groups sit between 17% and 28%: Telepathy is highest, 2.2 points over Freeze Time, and no group stands out. Segmented bars (Famous / Happy / Healthy / Rich, %): Fly 5.6 / 53.9 / 15.7 / 24.7 · Freeze Time 3.0 / 62.4 / 8.9 / 25.7 · Invisibility 3.1 / 63.3 / 11.2 / 22.4 · Super Strength 5.2 / 55.8 / 22.1 / 16.9 · Telepathy 0.0 / 72.1 / 0.0 / 27.9.
2	P(Fly | Famous) = 5/15 = 33.3% (plurality, not majority); P(Famous | Fly) = 5/89 = 5.6%. Spare pairs: 43/77 = 55.8% vs 43/265 = 16.2%; 63/101 = 62.4% vs 63/265 = 23.8%; 19/68 = 27.9% vs 19/102 = 18.6%.
3	P(Telepathy and Famous) = 0; P(Famous) = 15/433 = 3.5% vs P(Famous | Telepathy) = 0; P(F) × P(T) = 0.0054 vs 0. Independent 2x2: 60 sport, 50 instrument, 30 both (check 30/60 = 50/100). Mutually exclusive 2x2: 50 / 50 / 0 both (check 0/50 ≠ 50/100).
4	Regions 245 / 448 / 442 / 1,365. Union 1135/2500 = 45.4% (not 55.2%). P(internet | grad) = 245/693 = 35.4% vs P(internet) = 27.5% vs non-grads 442/1807 = 24.5%. The blog's 9.8% = 245/2500 is the joint. Product 0.0762 vs joint 0.098. Independence would predict 190.44 in both.
5	X: +8 (1/6), +1 (1/6), −2 (4/6). μ = 1/6 ≈ \$0.17 for the player; σ^2 = 485/36 ≈ 13.47, σ ≈ \$3.67. Fixes: 6 pays \$9, or 1 pays \$2. SD of a 20-game average ≈ \$0.82.
6	P(run of 6+ heads in 20 fair flips) = 0.1223 (exact value, by computer); 5+: 0.250; 7+: 0.058; 8+: 0.027.

Debrief notes

- **The two persistent rules:** the word after "given" names the group whose total is the denominator (posters 1 and 2); two events that can each happen and are mutually exclusive are never independent, because ruling one out changes the other's chance (poster 3, the 10/29 promise).
- **Card 1 nuance:** the newsletter's number is right and its reasoning is wrong. Accept "not supported" or "partly supported" when the sentence says the count is true and the percent does not stand out.
- **Card 5 nuance (only if an early finisher takes it):** fair means $\mu_X = 0$, for both sides; the question is who an unfair game favors. The 20-game averages will scatter; put them on one class dotplot and the law of large numbers teaches itself.
- **Cards 1 and 2, side by side:** 26/101 = 25.7% and 26/102 = 25.5% are nearly equal because the two given groups are nearly the same size. Flipping the given changes the answer a lot only when the two totals differ a lot: 5/15 against 5/89.
- **Card 6, Thursday and Friday:** the checkers' six trials are a second, smaller simulation. Accept the estimate as a fraction of 25, or of 31 if the six followed the same design and the group adds them in a second color. When the two estimates disagree, that shows a small simulation wobbles; it does not show which one is closer to the truth. That is the poster's sentence about what more trials do. The exact 0.1223 stays with you until Friday's debrief.
- **Not on the walls in Period B:** cards 4 and 5 (the double count, the fair game). Thursday's Do Now carries the double count; expected value waits for Friday's early finishers or for review.
- **Worksheet note (not for students):** the 2.7 follow-along's union examples print Super Status numbers (265 + 29 − 15; 115/433 + 29/433) that do not match the printed table (Super Strength is 77, Freeze Time 101, Happy and Super Strength 43). Every number in this kit uses the printed table; the worksheet blanks need a fix in a later pass.